

Surface Area and Volumes

Exercise = 10.3

Q1)- Find the Volume of the right circular cone with:

(i) Radius 3.5 cm, height 12 cm.

Sol. $r = 3.5 \text{ cm}$, $h = 12 \text{ cm}$

$$\text{Volume} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times (3.5)^2 \times 12$$

$$= \frac{22}{7} \times 12 \times 25 = \frac{22}{7} \times \left(\frac{35}{10}\right)^2$$

$$= \frac{22}{7} \times \frac{35}{10} = \frac{22}{7} \times 7 \times 7$$

$$= 22 \times 7$$

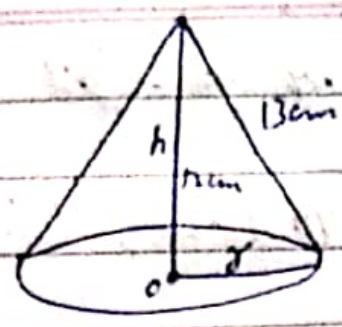
$$= 154 \text{ cm}^3.$$

Q2:- Find the capacity in litres of a conical vessel

with:

(i) Height 12 cm, slant height 13 cm.

Sol.



$$h = 12 \text{ cm}$$

$$l = 13 \text{ cm}$$

Since $l^2 = h^2 + r^2$

$$(13)^2 = (12)^2 + r^2$$

$$169 = 144 + r^2$$

$$\Rightarrow 169 - 144 = r^2$$

$$\Rightarrow r^2 = 25$$

$$\Rightarrow r = \sqrt{25}$$

$$\Rightarrow r = 5 \text{ cm}$$

$\therefore$ Volume of conical vessel.

$$= \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times (5)^2 \times 12$$

$$= \frac{22 \times 25 \times 4}{7} = \frac{2200 \text{ cm}^3}{7}$$

$$= \frac{2200}{7} \times \frac{1}{1000} \text{ l}$$

$$= \frac{22}{7} \times \frac{1}{10}$$

$$= \frac{11}{35} \text{ l.}$$

Q4: If the volume of a right circular cone of height 9cm is $48\pi \text{ cm}^3$, Find the diameter of its base.

Sol. $h = 9\text{ cm}$
Volume = $48\pi \text{ cm}^3$

$$\Rightarrow \frac{1}{3} \pi r^2 h = 48\pi$$

$$\frac{1}{3} \pi r^2 \times 9 = 48\pi$$

$$3r^2 = 48$$

$$r^2 = \frac{48}{3}$$

$$r^2 = 16$$

$$r = \sqrt{16}$$

$$r = 4\text{ cm}$$

$$\therefore \text{Diameter} = 2r$$

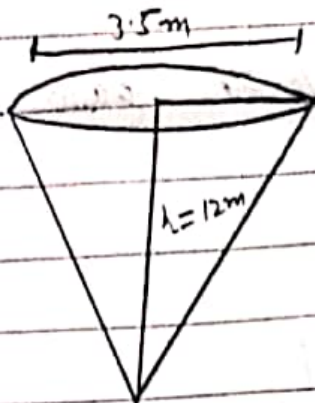
$$= 2 \times 4\text{ cm}$$

$$= 8\text{ cm.}$$

Q5: A conical pit of top diameter 3.5m is 12m deep

What is its capacity in kilolitres?

Soln



$$r = \frac{D}{2} = \frac{3.5 \text{ m}}{2}$$

$$= 1.75$$

$$h = 12 \text{ m}$$

i. Capacity (Volume) of conical pit = $\frac{1}{3} \pi r^2 h$

$$= \frac{1}{3} \times \frac{22}{7} \times (1.75)^2 \times 12$$

$$= \frac{88}{7} \times \left(\frac{357}{100} \right)^2 \text{ m}^3$$

$$= \frac{88}{7} \times \left(\frac{7}{4} \right)^2 \text{ m}^3$$

$$= \frac{88}{7} \times \frac{7 \times 7}{4 \times 4} \text{ m}^3$$

$$= \frac{77 \text{ m}^3}{2} = 38.5 \text{ m}^3$$

$$= 38.5 \text{ kl.}$$

$$- 1000 \text{ cm}^3 = 1 \text{ l}$$

$$1000 \times 1 \text{ cm} = 1 \text{ l}$$

$$\frac{1 \text{ cm}^3}{1000} = 1 \text{ l}$$

$$- 1 \text{ kl} = 1000 \text{ l}$$

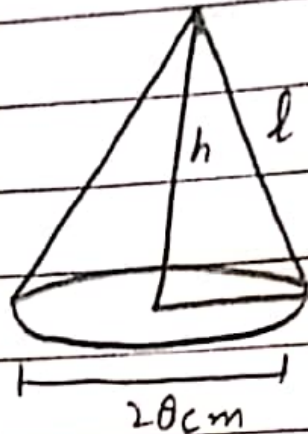
$$1000 \times 1 \text{ l} = 1 \text{ kl}$$

$$\frac{1 \text{ l}}{1000} = 1 \text{ kl}$$

$$- 1 \text{ m}^3 = 1000 \text{ l} / 1 \text{ kl}$$

Q6: The Volume of a right circular cone is 9856 m^3 .
If the diameter of the base is ~~28~~ 28 cm , find

(i) Height of the cone.



Sol Volume = 9856 cm^3

$$r = \frac{D}{2} = \frac{28}{2} \text{ cm}$$

$$r = 14 \text{ cm}$$

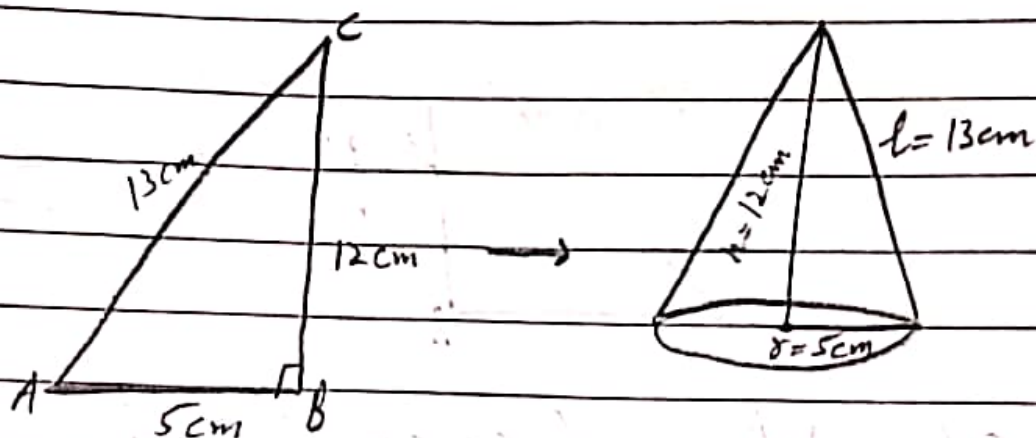
- Volume of rt. circular cone = 9856

$$\frac{1}{3} \pi r^2 h = 9856$$

$$\frac{1}{3} \times \frac{22}{7} \times (14)^2 \times h = 9856$$

$$\frac{1}{3} \times \frac{22}{7} \times 14 \times 14 \times h = 9856$$

Q7: A right triangle ABC with sides 5cm, 12cm and 13cm is revolved about the side 12cm. Find the volume of the solid so obtained.



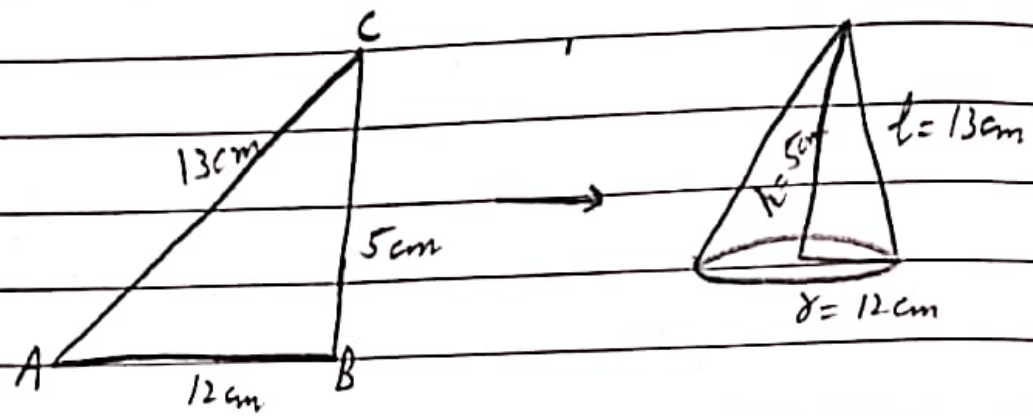
Sol: When we revolve right triangle ABC with respect to side 12cm. We obtain a rt. circular cone with $h = 12\text{cm}$, $r = 5\text{cm}$, $l = 13\text{cm}$

$$\therefore \text{Volume of Cone} = \frac{1}{3} \pi r^2 h = \frac{1}{3} \times \pi \times 5 \times 5 \times 12$$

$$= 100\pi \text{ cm}^3$$

Q8: If the triangle ABC in the Question 7 above is revolved about the side 5cm, then find the

volume of the solid ~~is~~ obtained. Find also the ratio of the volumes of the two solids obtained in Questions 7 and 8.



Sol: Volume of cone = $\frac{1}{3} \pi r^2 h$

$$= \frac{1}{3} \pi \times 12 \times 12 \times 5$$

$$= \pi \times 4 \times 12 \times 5$$

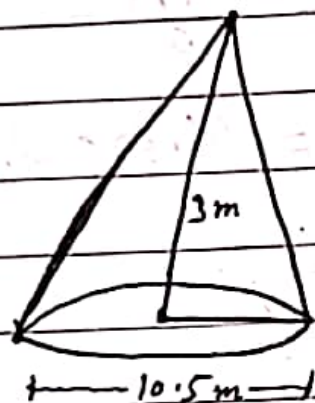
$$= 240\pi \text{ cm}^3$$

$\therefore$ Ratio = $\frac{\text{Volume of cone in Question 7}}{\text{Volume of cone in Question 8}}$

$$= \frac{1240 \pi \text{ cm}^3}{240 \pi \text{ cm}^3}$$

Ratio = 12:5

Q9: A heap of wheat is in the form of a cone whose diameter is 10.5 m and height is 3 m. Find its volume. The heap is to be covered by canvas to protect it from rain. Find the area of the canvas required.



Sol: Diameter of conical heap = 10.5 m

$$\therefore \text{Radius} = \frac{10.5}{2}$$

$$= 5.25$$

$$h = 3\text{m}$$

$$\therefore \text{Volume of conical heap} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times (5.25)^2 \times 3$$

$$= \frac{22}{7} \times 5.25 \times 5.25$$

$$= \frac{22}{7} \times \frac{525}{100} \times \frac{525}{100}$$

$$= \frac{22}{7} \times \frac{21}{4} \times \frac{21}{4}$$

$$= \frac{11 \times 3 \times 21}{8} = \frac{33 \times 21}{8} = \frac{693}{8}$$

$$= 86.625 \text{ m}^3$$

We know that. $l^2 = r^2 + h^2$

$$l^2 = (5.25)^2 + 3^2$$

$$l^2 = 27.56 + 9$$

$$l^2 = 36.56$$

$$\Rightarrow l = \sqrt{36.56}$$

$$\Rightarrow l = 6.05 \text{ m}$$

$\therefore$ Canvas required to cover the conical heap.

= ~~Canvas~~ Surface area of heap.
Covered

= T.S.A.

$$= \frac{22}{7} \times 5.25 \times 6.05 \text{ m}^2$$

$$= 99.825 \text{ m}^2$$